UNIT – III

Fracture & Fracture Mechanics

Subject : Mechanical Behaviour of Materials (23A03504c)

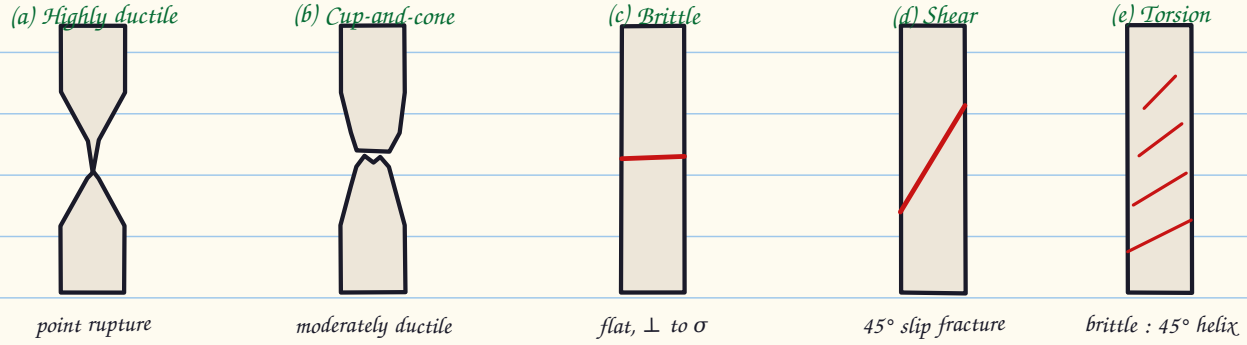
Syllabus: Types of fracture, mechanisms of ductile and brittle fracture, Griffith's theory, ductile-to-brittle transition temperature (DBTT) – factors & determination ; fracture mechanics – modes of fracture, stress intensity factor, strain energy release rate, fracture toughness and determination of K_{Ic}.

1. Introduction – Fracture

Fracture = separation of a body into two or more pieces under stress. Steps : (1) crack initiation (nucleation) and (2) crack propagation. Fracture is the end of useful life (and often catastrophic) – bridges, Liberty ships (1943–46 : ≈ 1500 cracks, 12 broke in two in cold water), Comet airliners (fatigue at window corners), pipelines, pressure vessels, Titanic (brittle steel, high S/P, cold sea).

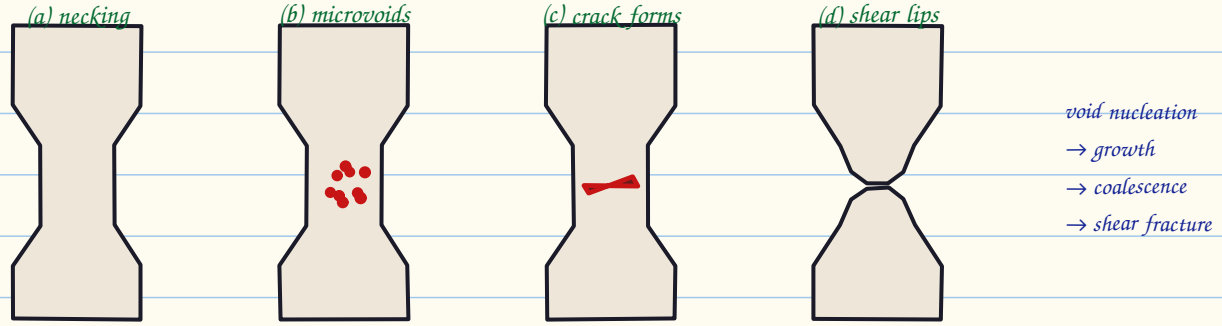
Classification of fracture

- **By ductility** : ductile (large plastic strain, slow stable crack, absorbs energy) / brittle (little plastic strain, fast unstable crack, flat surface) ; mixed (quasi-cleavage).
- **By crack path** : transgranular (through grains) / intergranular (along grain boundaries).
- **By loading** : tensile, shear, torsion, impact, **fatigue** (cyclic – Unit IV), **creep** (high T, time – Unit V), stress-corrosion, hydrogen-induced, thermal-shock fracture.



Feature	Ductile	Brittle
Plastic strain	large (> 5 % elongation)	negligible (< 1 %)
Crack speed	slow, needs increasing stress	fast, ≈ 0.4–0.5 × speed of sound
Energy absorbed	large	small
Warning	yes – visible necking	none
Surface	dull, fibrous, dimples	bright, granular / faceted, flat
Stress for propagation	increases with crack growth	may decrease (unstable)
Typical materials	Cu, Al, mild steel (> DBTT)	glass, ceramics, cast iron, BCC below DBTT

2. Ductile Fracture – Mechanism

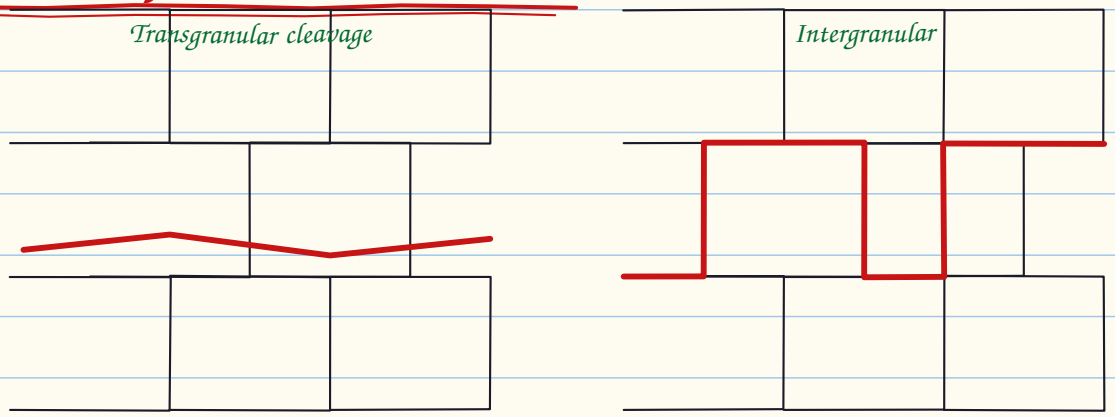


Cup-and-cone fracture of a round tensile bar (moderately ductile metals) :

- (a) After UTS a neck forms (plastic instability, $d\sigma = 0$) – local reduction of area, a **triaxial stress state** (hydrostatic tension) develops at the centre of the neck.
- (b) **Void (microcavity) nucleation** at inclusions (MnS, oxides), second-phase particles (carbides), by decohesion at the interface or particle cracking when local strain reaches a critical value.
- (c) **Void growth** (high triaxiality accelerates) and **coalescence** into an internal elliptical crack perpendicular to the axis at the centre of the neck ; ligaments between voids neck down (internal necking).
- (d) The crack grows outward until the outer rim, where the stress state is plane stress – it then **shears at ≈ 45°** (maximum shear stress) producing the **shear lip** (cone) ; the centre is flat, fibrous.

- **Fractography** : the centre shows equiaxed dimples (tensile), shear lip shows elongated (parabolic) dimples pointing to the origin ; dimple size depends on inclusion spacing – cleaner steels have deeper dimples and greater ductility.
- **Features** : high energy absorption, stable crack growth (needs increasing load), appears "fibrous". Very ductile metals (Au, Pb, Pt, soft Al) neck to a point (chisel/point rupture) ; ductile fracture in torsion is flat (shear), in thin sheets 45° slant fracture (single shear).
- Models : McClintock void growth ; Gurson–Tvergaard–Needleman (porous plasticity) ; critical void-volume fraction ≈ 0.15–0.25 for coalescence. **Ductility depends on** : inclusion content, stress triaxiality (notches reduce ductility), temperature, strain rate, texture.

3. Brittle Fracture – Mechanism



cracks cross grains on {100} planes

along weakened boundaries (temper embrittlement, H, S)

- Occurs with little or no plastic deformation, by rapid crack propagation perpendicular to the maximum tensile stress ; the crack runs at a high fraction of the speed of sound – failure without warning.
- **Cleavage** (transgranular) : separation along specific low-index crystal planes – {100} in BCC (Fe, W, Mo), {0001} in HCP (Zn, Mg), {111} in diamond cubic ; facets of cleavage show "river patterns" (steps merging in direction of crack growth) and **chevron / herringbone marks** on large fractures – the V's point back to the origin.
- **Intergranular** : crack path along grain boundaries – due to weak boundaries : grain-boundary films (e.g. cementite networks, S/P segregation), temper embrittlement (Sb, Sn, P segregation at 350–550 °C), hydrogen embrittlement,

liquid-metal embrittlement, stress-corrosion cracking, creep cavitation. Shiny, rock-candy appearance.

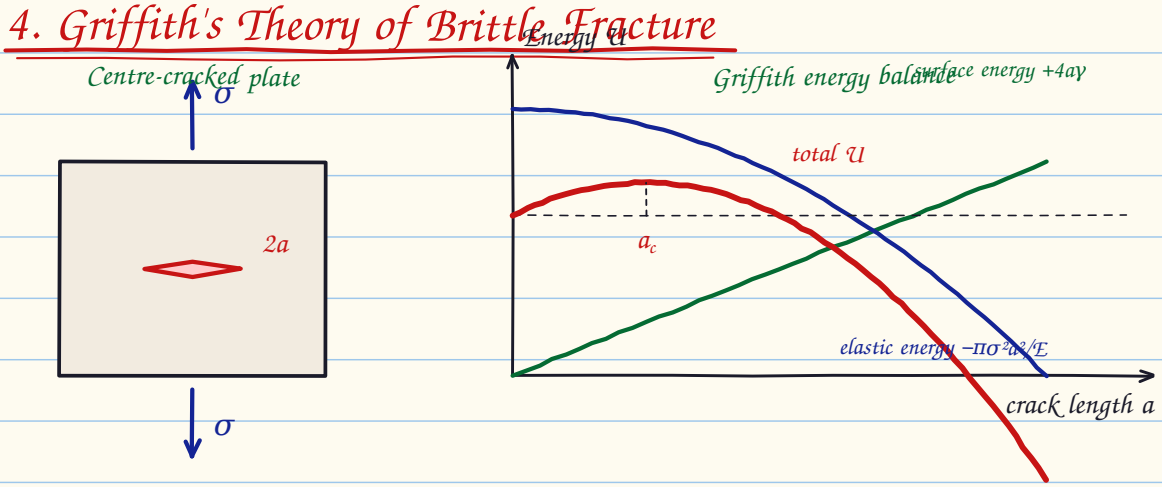
- **Mechanism of cleavage-crack nucleation** : dislocation pile-up of n dislocations on a slip plane against a grain boundary or a carbide ⇒ stress concentration $n\,\tau$; two interacting dislocations coalesce into a wedge crack nucleus (Cottrell : a/2[111] + a/2[1-1-1] → a[100]) ; or twin intersection ; or a cracked carbide particle (steels).

$$(\tau - \tau_i)\, k_y\, d^{1/2} \geq \beta\, G\, \gamma_{eff} \quad \text{(Stroh–Cottrell criterion)} \quad \sigma_f = \sigma_i + k_f\, d^{-1/2}$$

- Brittle fracture is favoured by : (a) triaxial stress (notch, thick section), (b) low temperature, (c) high strain rate (impact), (d) coarse grains, (e) BCC/HCP crystal structure, (f) hardened or irradiated steel. (The "three factors" of Orowan.)
- **Stress concentration** (Inglis) : $\sigma_{max} = \sigma[1 + 2\sqrt{(a/\rho)}] = K_t\,\sigma$ (a = semi-axis perpendicular to the load, ρ = tip radius) ; sharp cracks $\rho \to 0$ ⇒ stress → ∞ ⇒ need the energy criterion.

Ex 1. Elliptical hole a = 5 mm, ρ = 0.5 mm : $K_t = 1 + 2\sqrt{(5/0.5)} = 7.3$ ⇒ local stress 7.3 σ.

4. Griffith's Theory of Brittle Fracture



Observed fracture strength of brittle solids (glass ≈ 70–100 MPa) is far below the theoretical cohesive strength $\sigma_{th} \approx \sqrt{(E\gamma_s/a_0)} \approx E/10$ (≈ 8 GPa). **Griffith (1920)** : the solid contains pre-existing microcracks that raise the local stress ; a crack extends if the **elastic strain energy released** is at least equal to the **surface energy required** to create the new surfaces.

Energy balance for a through-crack of length 2a (unit thickness) in a large plate (plane

- Elastic energy released (stress relieved in a region $\approx \pi a^2$) : $\mathcal{U}_e = -\pi \sigma^2 a^2 / E$.
- Surface energy needed : $\mathcal{U}_s = 2\,(2a)\,\gamma_s = 4\,a\,\gamma_s$ (two faces).
- Total $\mathcal{U} = 4a\gamma_s - \pi\sigma^2 a^2/E$. The crack becomes unstable when $d\mathcal{U}/da = 0$:

$$4\gamma_s = 2\pi\sigma^2 a/E \Rightarrow \sigma_f = \sqrt{(\,2E\gamma_s / \pi\, a\,)} \text{ (plane strain : } E \to E/(1-\nu^2))$$

- Interpretation : for a given stress σ the critical crack length $a_c = 2E\gamma_s/(\pi\sigma^2)$; $\mathcal{U}$ reaches a maximum at a_c – shorter cracks close, longer cracks grow spontaneously (energy decreases). Fracture stress $\propto 1/\sqrt{a}$ – larger flaws ⇒ lower strength (strength scatter in ceramics ; size effect ; fibres).

Ex 2. Glass $E = 70$ GPa, $\gamma_s = 0.3$ J/m², flaw $a = 1$ µm : $\sigma_f = \sqrt{(2 \times 70\times10^9 \times 0.3/(\pi \times 10^{-6}))} = 116$ MPa ✓ (observed). For $a = 1$ nm (atomic flaw) $\sigma \approx 3.7$ GPa — close to the theoretical strength.

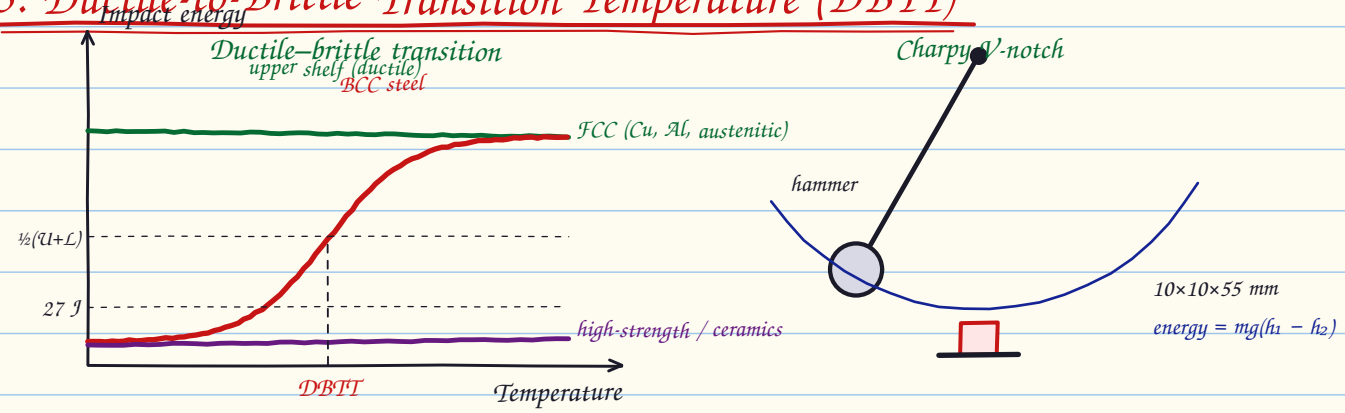
Orowan / Irwin modification for ductile metals

- Plastic work γ_p at the crack tip (10^3–$10^4 \times \gamma_s$ for steel) dominates :

$$\sigma_f = \sqrt{(\,2E\,(\gamma_s + \gamma_p)\,/\,\pi\, a\,)} \approx \sqrt{(\,2E\gamma_p\,/\,\pi\, a\,)}$$

- Griffith's theory is accurate for glass, ceramics, ice, silicon, polymers below T_g ; with the plastic-zone correction it works for metals (LEFM). **Limitations** : needs the pre-existing crack, ignores plastic zone & crack tip geometry.
- Hence strength of brittle materials is governed by flaw size ⇒ *surface finish, fire-polishing, tempering (compressive surface stress)* raise the strength of glass.

5. Ductile-to-Brittle Transition Temperature (DBTT)



Many materials, especially BCC metals (mild steel, Fe, Mo, W, Cr, Nb), fail by ductile fracture above a critical temperature and by brittle cleavage below it – the **ductile–brittle transition.** FCC metals (Cu, Al, Ni, austenitic steel) and very high-strength alloys do not show a sharp transition.

- **Cause** : the yield stress of BCC metals rises rapidly as temperature falls (high Peierls stress of screw dislocations) while the cleavage fracture stress is nearly temperature-independent. Below T_c the cleavage stress is reached before yielding ⇒ brittle ; Petch : $\beta\ T_c = \ln B - \ln(C\ d^{1/2})$ (finer grains lower T_c).
- **Charpy curve regions** : **upper shelf** – ductile (high energy, 100 % fibrous) ; **transition region** – mixed, large scatter ; **lower shelf** – brittle (≈ 5–20 J, 100 % cleavage).

Factors affecting DBTT

Factor	Effect	Direction
Carbon	≈ +14 °C per 0.1 % C ; pearlite (↑)	raises
Manganese	Mn : C > 3 ⇒ lowers (≈ −5 °C per 0.1 %), refines grain	lowers
Nickel	−10 to −20 °C per 1 % Ni ; 9 % Ni for LNG tanks (−196 °C)	lowers
P, S, O, N, Sb, Sn	segregation / inclusions / strain ageing	raise
Grain size	finer grain ⇒ lower DBTT (Petch)	finer lowers
Microstructure	tempered martensite (lowest) < bainite < pearlite+ferrite < coarse pearlite	—
Strain rate / temperature	high rate (impact) shifts curve to higher T	rate raises
Notch / triaxiality / thickness	sharp notch, thick plate, residual & welding stresses	raise
Cold work, hardening, irradiation	↑ yield, loss of toughness (neutron damage in reactor vessels +50–150 °C)	raise
Si, Al-killed practice	fully killed, normalised steel has lower DBTT	killing lowers

- Remedies : fine-grained killed Al steels, Ni additions, normalising/quench-tempering, controlled rolling (TMCP), avoid low-temperature

service or use FCC (austenitic stainless, Al, Cu), keep S & P low (< 0.01 %).

Determination of DBTT

- **Charpy V-notch impact test** (ASTM E23) : specimen 10 × 10 × 55 mm, 2 mm deep 45° notch, simply-supported, struck by a pendulum – energy absorbed = $mg(h_1 - h_2)$; tests at a series of temperatures (−196 → +100 °C, baths of liquid N_2, alcohol-dry ice, ice, hot water). **Izod** : cantilever mounting. Instrumented Charpy gives load–time curve.
- Criteria for DBTT (from the energy–temperature curve) : (1) mid-point of upper and lower shelf energy ; (2) fixed energy level – 20 J (15 ft-lb) for ships, 27 J (20 ft-lb) for structural steel ; (3) FATT – 50 % fibrous (shear) fracture appearance ; (4) lateral expansion 0.38 mm (15 mil) ; (5) NDT temperature.
- **Nil-ductility transition (NDT)** temperature – drop-weight test (Pellini, ASTM E208) : the highest T at which a specimen with a brittle weld bead breaks. Pellini's **fracture analysis diagram** : NDT, FTE (NDT + 33 °C : crack arrest), FTP (NDT + 67 °C : plastic overload) – used to choose service temperature (service T > FTE).

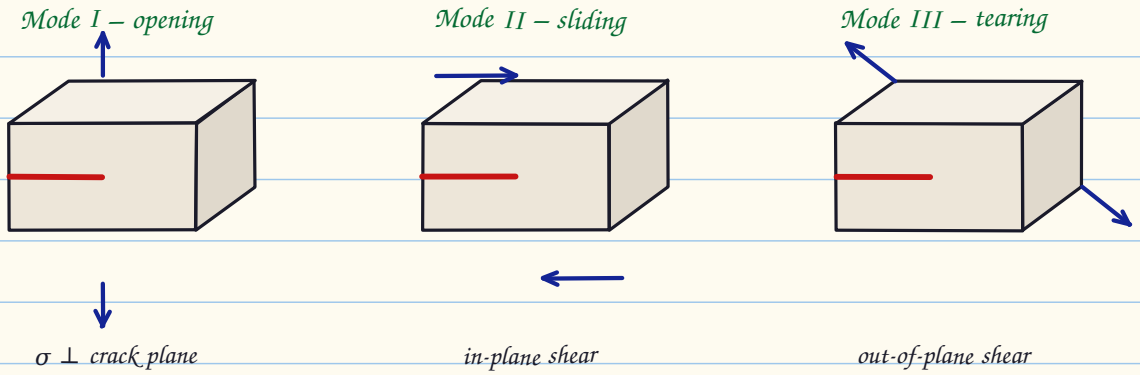
> Ex 3. Steel energy–temperature data : upper shelf 160 J, lower shelf 10 J ⇒ mid-point energy = 85 J ; reading the temperature at 85 J gives the DBTT ; the 27 J temperature is about 25 °C lower — specification may require 27 J at −20 °C.

6. Fracture Mechanics – Introduction

Conventional design uses $\sigma < \sigma_y/FS$; it ignores flaws. Fracture mechanics relates the applied stress, flaw size and material toughness – allows computing the safe stress, critical flaw size and inspection intervals (damage-tolerant design).

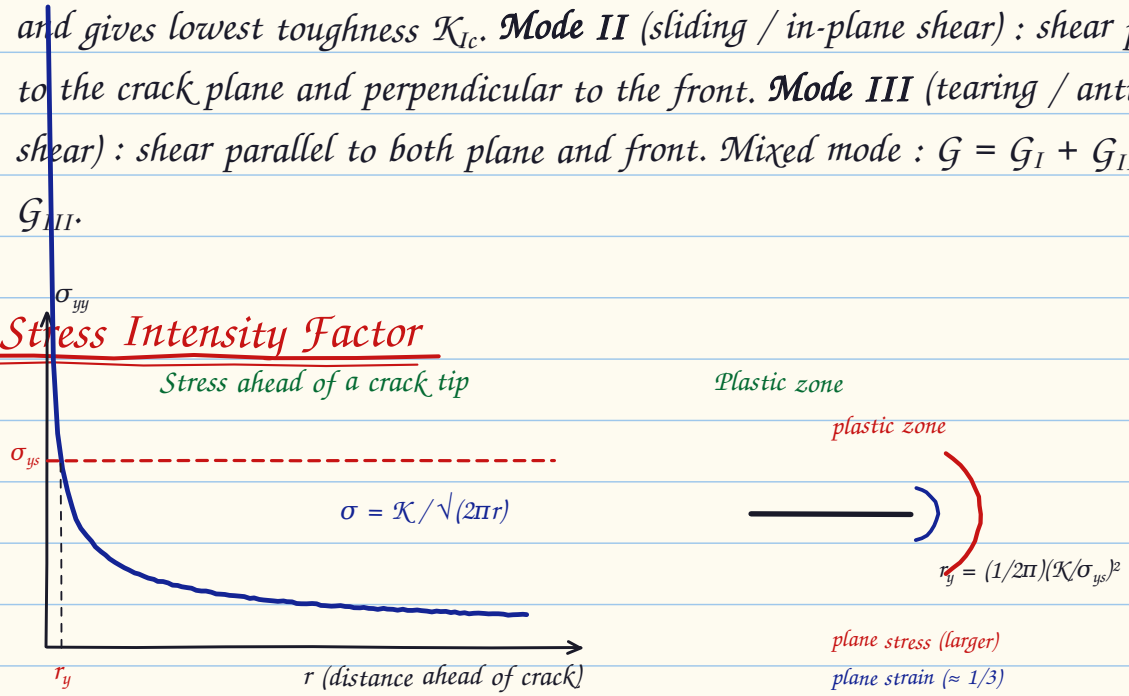
- **LEFM** (linear elastic fracture mechanics) : the plastic zone is small compared with crack length and specimen size. **EPFM** (elastic-plastic) : J-integral, CTOD for tough materials.

Modes of crack loading



- **Mode I** (opening) : tensile stress normal to the crack plane – the most important, and gives lowest toughness K_{Ic}. **Mode II** (sliding / in-plane shear) : shear parallel to the crack plane and perpendicular to the front. **Mode III** (tearing / anti-plane shear) : shear parallel to both plane and front. Mixed mode : $G = G_I + G_{II} + G_{III}$.

7. Stress Intensity Factor



Near the tip of a sharp crack ($r << a$) in a linear elastic body :

$$\sigma_{ij} = (K_I / \sqrt{(2\pi\ r)}) f_{ij}(\theta)\ K_I = Y \sigma \sqrt{(\pi\ a)}\ (MPa\sqrt{m})$$

- K_I is the **stress intensity factor** – a single parameter characterising the whole crack-tip field (stress $\propto K/\sqrt{r}$, singular) ; depends on applied stress σ, crack length a and geometry factor Y. Ahead of the crack on the plane $\theta = 0$: $\sigma_{yy} = K_I/\sqrt{(2\pi r)}$.
- Y : centre crack in an infinite plate 1.0 (a = half length) ; single edge crack 1.12 ($1.12\sqrt{\pi} = 1.99$) ; double edge crack ≈ 1.12 ; embedded penny crack $2/\pi$; surface half-circular crack 0.71 ; for finite width plates $Y = \sqrt{(\sec(\pi a/W))}$ (centre crack).
- **Fracture criterion** : unstable fracture occurs when $K_I = K_{Ic}$ (the plane-strain fracture toughness, a material property ; $MPa\sqrt{m}$). Safe if $K_I < K_{Ic}$.

$$\sigma_c = K_{Ic} / (Y\sqrt{(\pi\ a)})\ a_c = (1/\pi)\ [\ K_{Ic} / (Y\ \sigma)\ ]^2$$

Material	K_{Ic} (MPa$\sqrt{m}$)
Mild steel	140
4340 steel (σ_y 1500 MPa)	50
Maraging steel	90–110
Al 2024-T3 / 7075-T6	34 / 24–30
Ti-6Al-4V	55–115
Cast iron	6–20
Alumina / SiC	3–5 / 3–4
Glass	0.7
PMMA	1.0
Concrete	0.2–1.4

Ex 4. Plate with an edge crack, K_{Ic} = 50 MPa$\sqrt{m}$, σ = 300 MPa, Y = 1.12 : a_c = $(1/\pi)(50/(1.12\times300))^2$ = $(1/\pi)(0.1488)^2$ = 7.05 mm. If the largest crack found by inspection is a = 2 mm and K_{Ic} = 60 MPa$\sqrt{m}$: σ_c = $60/(1.12\sqrt{(\pi\times0.002)})$ = 676 MPa.

Plastic zone & small-scale yielding

- The elastic stress singularity is truncated by yielding : plastic-zone radius (Irwin)

$$r_y = (1/2\pi)(K_I/\sigma_{ys})^2 \text{ (plane stress)}\quad r_y = (1/6\pi)(K_I/\sigma_{ys})^2 \text{ (plane strain)}$$

- Effective crack length $a_{eff} = a + r_y$ (Irwin correction) – valid when $r_y < a/50$. Plane strain (thick plate) : triaxial constraint, small zone, flat fracture, lowest K_c ⇒ K_{Ic} ; plane stress (thin sheet) : larger zone, slant fracture, K_c higher.

Ex 5. K = 50 MPa$\sqrt{m}$, σ_{ys} = 1000 MPa : plane stress r_y = $(1/2\pi)(0.05)^2$ = 0.40 mm ; plane strain = 0.13 mm.

8. Strain Energy Release Rate

Energy released per unit crack-extension area (per unit thickness per unit crack advance) :

$$G = -(1/B)\ \partial U_e/\partial a = \pi\ \sigma^2\ a\ /\ E \text{ (plane stress)} ;\ G = \pi\ \sigma^2\ a\ (1-\nu^2)\ /\ E \text{ (plane strain)}$$

$$G_I = K_I^2 / E \quad E = E \text{ (plane stress)} ; E = E/(1 - \nu^2) \text{ (plane strain)}$$

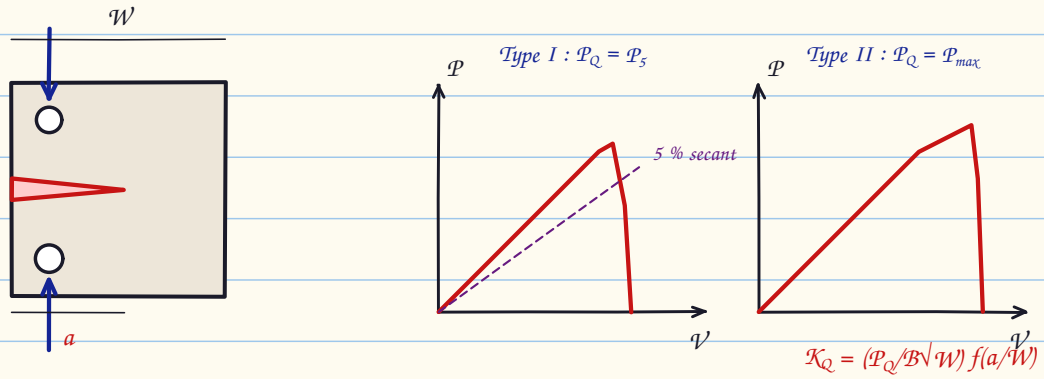
- Units J/m^2 (N/m). Crack growth when $G \geq G_c$ (critical strain energy release rate = energy needed to create new fracture surface $\approx 2\gamma_s + 2\gamma_p$) ; $G_{Ic} = K_{Ic}^2(1 - \nu^2)/E$. Equivalence of Griffith and Irwin : $K_I^2 = E\,G$.
- Griffith's relation from G : $G_c = 2\gamma_s \Rightarrow \sigma_f = \sqrt{(2E\gamma_s/\pi a)}$ (ideally brittle). ***Load control vs displacement control*** : in fixed-grip loading, G decreases with a (stable crack growth) ; constant load – G increases with a (unstable).
- ***R-curve*** : crack-growth resistance versus crack extension ; flat $R = G_c$ (plane strain), rising for thin sheets (plane stress) ; instability when $dG/da > dR/da$.

Ex 6. $K = 50\ MPa\sqrt{m}$, $E = 207\ GPa$: $G = K^2/E = (50\times10^6)^2/(207\times10^9) = 12.1\ kJ/m^2$ (plane stress) ; plane strain $\times(1 - 0.3^2) = 11.0\ kJ/m^2$.

- ***EPFM parameters*** : J-integral (energy flux, path independent) – $J = G$ in linear elasticity, J_{Ic} ; CTOD (crack-tip opening displacement) $\delta = K^2/(\sigma_{ys}E) = G/\sigma_{ys}$; for tough low-strength steels (ASTM E1820).

9. Fracture Toughness Testing – Determination of K_{Ic}

Compact tension (CT) Load–displacement records (E399)



ASTM E399 (plane-strain K_{Ic} testing) : specimens – compact tension (CT), single-edge notch bend (SENB, three-point), arc-shaped, disc compact.

- ***Specimen*** : CT with width $W = 2B$, $a/W = 0.45$–0.55 ; machined notch + fatigue pre-crack (sharp crack, $K_{max} \leq 0.6\ K_{Ic}$) of ≈ 1.3 mm.
- ***Test*** : load at slow rate (loading rate $dK/dt = 0.55$–$2.75\ MPa\sqrt{m}/s$) in a universal testing machine, record load P versus clip-gauge opening V (COD). Draw a secant line at 95 % of the initial slope (5 % secant) – intersection gives P_5 ; $P_Q = P_5$ if load before P_5 is lower (curve type I), or maximum load before P_5

(type II / III).

- Compute the conditional value

$$K_Q = (P_Q / (B\sqrt{W})) f(a/W)\, f(\alpha) = (2+\alpha)(0.886 + 4.64\alpha - 13.32\alpha^2 + 14.72\alpha^3 - 5.6\alpha^4) / (1-\alpha)^{3/2}$$

- **Validity (plane-strain) conditions** : $P_{max}/P_Q \le 1.10$ and

$$B,\ a,\ (W-a) \ge 2.5\,(K_Q/\sigma_{ys})^2$$

If valid, $K_Q = K_{Ic}$; else a thicker specimen is needed (K_c depends on thickness – plateau at $B \ge 2.5(K_{Ic}/\sigma_{ys})^2$).

Ex 7. CT : $B = 25$ mm, $W = 50$ mm, $a = 25$ mm ($\alpha = 0.5 \Rightarrow f = 9.66$), $P_Q = 20$ kN, $\sigma_{ys} = 500$ MPa : $K_Q = (20\times10^3/(0.025 \times \sqrt{0.05})) \times 9.66 = 3.58\ \text{MPa}\sqrt{\text{m}} \times 9.66 = 34.6\ \text{MPa}\sqrt{\text{m}}$. Check : $2.5(34.6/500)^2 = 0.0120$ m $= 12$ mm < 25 mm ✓ $\Rightarrow$ $K_{Ic} = 34.6\ \text{MPa}\sqrt{\text{m}}$. Required B for a 1000 MPa steel with $K_{Ic} = 50$: $B \ge 2.5(50/1000)^2 = 6.25$ mm.

- Other tests : SENB ($K = (P\,S/(B\,W^{3/2}))\, f(a/W)$) ; chevron-notch for ceramics (ASTM C1421) ; indentation fracture (ceramics) $K_c \approx 0.016\,(E/H)^{1/2}\,P/c^{3/2}$; J_{Ic} (ASTM E1820), CTOD (BS 7448), crack-arrest K_{Ia}.
- Uses : material selection (K_{Ic}/σ_{ys} ratio), critical flaw size, NDT inspection limits, leak-before-break design, remaining-life estimation. Increase of toughness : decrease strength, refine grain, clean steel, avoid embrittling phases, tempering ; metals with high K_{Ic}/σ_{ys} are fracture-tolerant.

10. Additional Solved Problems

P1. A tensile bar with a circumferential flaw $a = 1.5$ mm, $Y = 1.12$, $K_{Ic} = 40\ \text{MPa}\sqrt{\text{m}}$: $\sigma_c = 40/(1.12\sqrt{(\pi \times 0.0015)}) = 40/0.0769 = 520$ MPa.

P2. Ceramic ($E = 380$ GPa, $\gamma_s = 1\ \text{J/m}^2$, flaw $a = 20\ \mu$m) : $\sigma_f = \sqrt{(2\times380\times10^9\times1/(\pi\times20\times10^{-6}))} = 110$ MPa.

P3. G for $\sigma = 200$ MPa, $a = 10$ mm, $E = 200$ GPa : $G = \pi(200\times10^6)^2(0.01)/200\times10^9 = 6.3\ \text{kJ/m}^2 \Rightarrow K = \sqrt{(EG)} = 35.5\ \text{MPa}\sqrt{\text{m}}$ ($= \sigma\sqrt{(\pi a)} = 200 \times 0.1772\ \sqrt{}$).

P4. Which is safer, steel A (σ_{ys} 1500, K_{Ic} 60) or B (σ_{ys} 1000, K_{Ic} 100)? Critical a at $\sigma = 0.5\sigma_{ys}$ ($Y = 1.12$) : A : $(1/\pi)(60/(1.12\times750))^2 = 1.6$ mm ; B : $(1/\pi)(100/(1.12\times500))^2 = 10.2$ mm – steel B is far more fracture-tolerant.

Short answers – Quick Revision

- *Why is a brittle fracture dangerous?* No warning, rapid unstable growth, low energy absorption.
- *Why do FCC metals not show DBTT?* Yield stress is only weakly temperature dependent – dislocations stay mobile at low T.
- *Why is K_{Ic} measured under plane strain?* It is the lowest, conservative, thickness-independent toughness.
- *Relation between K and G?* $G = K^2/E$ ($E = E$ or $E/(1-\nu^2)$).
- *Use of Charpy test?* Qualitative ranking of notch toughness and DBTT, quality control ; not a design property (no direct link to K_{Ic}, but Barsom–Rolfe correlation).

11. Quick Formula Sheet – Unit III

Item	Formula
Stress concentration	$\sigma_{max} = \sigma(1 + 2\sqrt{(a/\rho)})$
Griffith	$\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$; with plasticity $\sqrt{(2E(\gamma_s+\gamma_p)/\pi a)}$
Stress intensity	$K_I = Y\sigma\sqrt{(\pi a)}$; fracture if $K_I \geq K_{Ic}$
Critical crack	$a_c = (1/\pi)(K_{Ic}/Y\sigma)^2$
Plastic zone	$r_y = (1/2\pi)(K/\sigma_{ys})^2$; plane strain ÷3
Energy release rate	$G = K^2/E = \pi\sigma^2 a/E$
Validity (E399)	$B, a, W-a \geq 2.5(K_Q/\sigma_{ys})^2$; $P_{max}/P_Q \leq 1.1$

12. Important Questions

- Differentiate between ductile and brittle fracture; explain the cup-and-cone fracture mechanism.
- Explain the mechanism of cleavage fracture ; transgranular vs intergranular.
- Derive Griffith's criterion for brittle fracture ; state its modification for metals.

- What is DBTT? Discuss the factors affecting it and the methods of its determination (Charpy, NDT, FATT).
- Define modes of fracture, stress intensity factor, strain energy release rate ; derive the relation between G and K.
- Describe the determination of K_{Ic} (ASTM E399) with specimen, procedure and validity criteria.